DR. VARDAN ARUTIUNIAN, PH.D.

CURRICULUM VITAE

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RESEARCH PROGRAM

I am a developmental cognitive neuroscientist investigating the neural mechanisms underlying cognition, perception, communication, and adaptive behavior across typical and atypical development. My research integrates electrophysiology (EEG/MEG), genomics, artificial intelligence, and clinical trials to investigate brain function across multiple levels of analysis, from genes and neural systems to cognition and behavior. Working at the interface of psychology, neuroscience, genetics, and clinical research, I combine fundamental discovery with translational research to advance our understanding of human cognition while developing objective biomarkers that improve diagnosis, prognosis, intervention, and health outcomes for individuals with neurodevelopmental disorders.

ACADEMIC POSITIONS

2025 – present	**Postdoctoral Fellow,** Neuroscience of Early Development Lab, CHU Sainte-Justine (Mother and Child University Hospital Center), University of Montreal, Montreal, QC, Canada
2022 – 2025	**Postdoctoral Fellow,** Psychophysiology and Behavioral Systems Lab, Center for Child Health, Behavior and Development, Seattle Children's Research Institute, Seattle, WA, United States of America
2018 – 2022	**Junior Research Fellow,** Center for Language and Brain, National Research University Higher School of Economics (HSE University), Moscow, Russia
2012 – 2017	**Research Assistant,** Institute of Humanities, Immanuel Kant Baltic Federal University, Kaliningrad, Russia

EDUCATION

2018 – 2022	Ph.D. in (Neuro)Linguistics, *with cum laude* PhD dissertation: *Language Impairment in Children with Autism Spectrum Disorder: Linguistic Aspects* National Research University Higher School of Economics (HSE University), Moscow, Russia
2007 – 2012	B.A. & MSc. (undergraduate program) in Linguistics Immanuel Kant Baltic Federal University, Kaliningrad, Russia

AWARDS AND GRANTS

Pending	Simons Foundation: project *Excitation/inhibition imbalance as a neural marker of language impairment in autism: Genetic and neurophysiological approaches to define different language subgroups;* **principal investigator**
Pending	Autism Research Institute, USA: project *Sustained auditory brain activity links genomic variation to language functioning in children with autism: A family-based analysis;* **principal investigator**
Pending	Canadian Institutes of Health Research, Canada: project *A Family-Based Investigation of the Heritability of Neural Mechanisms Underlying Sensory Processing in Neurodevelopmental Disorders;* **co-investigator**
2026	Research Network on the Developmental and Intergenerational Origins of Children's Health (Réseau ODISÉ), travel grant for postdoctoral fellows.
2025 – 2026	Autism Research Institute, USA: project *Excitation/inhibition imbalance as a neural marker of language impairment in youth with Autism Spectrum Disorder (ASD): Relation to sex and lan-*

guage phenotype; **principal investigator** ($50,000)

2025 Hearst Fellowship, travel grant for postdoctoral fellows: Seattle Children's Research Institute, USA; *genetic and genomic data analysis at Yale School of Medicine, Yale University* ($3,000)

2024 – 2025 Simons Foundation, USA: project *Excitation/inhibition imbalance, its neural correlates and relation to clinical phenotypes in youth with Autism Spectrum Disorder*; **mentor** ($11,500)

2024 Ventura Endowed Fellowship Award: University of Washington/Seattle Children's Research Institute, USA; *Clinical training in ADOS administration* ($2,000)

2023 – 2024 Simons Foundation, USA: project *Early predictors of language impairment in infants at risk for developing autism*; **mentor** ($28,074)

2022 Russian Science Foundation: project *The development of auditory gamma oscillations in children, their relation to age, language abilities, and non-verbal intelligence: A magnetoencephalography (MEG) study;* **principal investigator** ($17,000)

2021 Best Lecturer – 2021, HSE University ($3,300)

2020 International Brain Research Organization (IBRO): InEurope Short Stay Grants Program for PhD Students and Post-Docs (€3,000)

2019 – 2021 The award from the Government of Russian Federation for PhD students ($3,500)

2019 – 2020 Young Faculty Support Program (Group of Young Academic Professionals). Category *New Researchers,* HSE University ($12,000)

2019 – 2020 Russian Foundation for Basic Research (RFBR): project *Reading acquisition in Russian-speaking school students: an eye-tracking study of linguistic processing contribution to mastering reading skills*; **contributor** ($139,000)

2018 – 2022 Russian Federation Government Grant № 14.641.31.0004: project *Language and Brain: diagnostics and correction of language impairments*; **contributor** ($2,700,000)

2018 – 2021 Russian Foundation for Basic Research (RFBR): project *Linguistic deficit and its brain correlates in children with dyslexia*; **contributor** ($86,000)

2018 – 2019 Russian Foundation for Basic Research (RFBR): project *Single-word, sentence and discourse comprehension in patients with temporal lobe epilepsy (including effect of non-verbal IQ and verbal working memory span)*; **contributor** ($12,800)

2016 – 2018 Russian Foundation for Basic Research (RFBR): project *Neurocognitive mechanisms for reading: An orthographical processing in the light of information theory by Claude Shannon (Evidence from eye movements)*; **principal investigator** ($14,300)

PUBLICATIONS

Articles in international peer-reviewed journals (published)

1. **Arutiunian, V.,** Davoudi, S., Gaougaou, G., Knoth, I.S., Buckser, R., Marcil, V., & Lippé, S. (2026). Auditory sustained potential as a biomarker of language functioning in children with autism and its implication in clinical trials. *Frontiers in Psychiatry, 17,* 1892698.
2. **Arutiunian, V.**, Sugar, C.A., Naples, A., Santhosh, M., Levin, A.R., Borland, H., Benton, J., Bernier, R.A., Chawarska, K., Dawson, G., Dziura, J., Hellemann, G., Faja, S., Jeste, S., Kleinhans, N., Sabatos-DeVito, M., Şentürk, D., Shic, F., McPartland. J.C., & Webb, S.J. (2026). EEG Resting-State Gamma Power as a Potential Biomarker of Excitation/Inhibition Imbalance in Autism: Evidence from the Autism Biomarkers Consortium for Clinical Trials. *European Neuropsychopharmacology, 113,* 113864.
3. **Arutiunian, V.,** Santhosh, M., Neuhaus, E., Borland, H., Bernier, R.A., Bookheimer, S.Y., Dapretto, M., Gupta, A.R., Jack, A., Jeste, S., McPartland, J.C., Naples, A., van Horn, J.D., Pelphrey, K.A., & Webb, S.J. (2026). Altered aperiodic EEG spectral power during speech perception task is associated with verbal communication in youths with Autism Spectrum Disorder. *Scientific Reports, 16,* 28881.
4. **Arutiunian, V.**, Sullivan, C.A.W., Santhosh, M., Neuhaus, E., Borland, H., Bernier, R.A., Bookheimer, S.Y., Dapretto, M., Jack, A., Jeste, S., McPartland, J.C., Naples, A., van Horn, J.D., Pelphrey, K.A., Webb, S.J., Gupta, A.R. (2026). Number of Alpha Peaks in the Electroencephalogram Is Associated With Clinical Phenotype and Copy Number Variants in Youths With Autism. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging, 11,* 300–309.

5. **Arutiunian, V.**, Sullivan, C.A.W., Santhosh, M., Neuhaus, E., Borland, H., Bernier, R.A., Bookheimer, S.Y., Dapretto, M., Jack, A., Jeste, S., McPartland, J.C., Naples, A., van Horn, J.D., Pelphrey, K.A., Webb, S.J., Gupta, A.R. (2026). Excitation/inhibition balance subtypes in autism and their genetic, neural, and clinical profiles. *Journal of Neural Transmission,* 1–16.
6. Novoselova, K., Lopukhina, A., Gomozova, M., Fedorov, M., Davydova, E., Pereverzeva, D., Sorokin, A., Tyushkevich, S., Mamokhina, U., Danilina, Dragoy, O., & **Arutiunian, V.** (2026). Difference in Language Profiles of Children With Autism Spectrum Disorder and Down Syndrome Is Not Driven by Non-Verbal Cognition. *International Journal of Language and Communication Disorders, 61,* e70177.
7. Minnigulova, A., Dragoy, O., & **Arutiunian, V.** (2026). Contribution of attentional mechanisms to verbal and nonverbal communication in boys with ASD. *European Child and Adolescent Psychiatry, 35,* 3239–3249.
8. Minnigulova, A., Protopova, M., Dragoy, O., & **Arutiunian, V.** (2026). Atypical Social Behavior is Predicted by Overconnectivity Between Salience and Default Mode Networks in Autism Spectrum Disorder. *Journal of Autism and Developmental Disorders, 56,* 3844–3852.
9. Rea, H., Mattson, J.T., **Arutiunian, V.,** Verdes, A., Bhattachan, S., Garcia, A., Ajwani, A., Dyer, M., Faja, S., Hanson, E., Pawlowski, K., Goin-Kochel, R.P., Prok, L., Veenstra-VanderWeele, J., Walton-Bowen, K., Webb, S.J., & Wang, P.P. (2026). Rationale and study design for the first precision medicine randomized placebo-controlled trial in the 16p11.2 deletion syndrome. *Expert Review of Neurotherapeutics, 26,* 597–607.
10. Samoylov, I., Bolgina, T., Lonshakov, G., Scheglova, E., Gomozova, M., Yashina, T., **Arutiunian, V.,** & Dragoy, O. (2026). Atypical neural activation and age-related patterns in dyslexia. *Cerebral Cortex, 36,* bhag122.
11. **Arutiunian, V.**, Buyanova, I., Minnigulova, A., Davydova, E., Pereverzeva, D., Sorokin, A., Tyushkevich, S., Mamokhina, U., Danilina, K., & Dragoy, O. (2025). Left-hemispheric atypicalities in the primary auditory cortex are associated with language comprehension and social skills in school-aged children with Autism Spectrum Disorder. *Cerebral Cortex, 35(3),* bhaf055.
12. **Arutiunian, V.**, Santhosh, M., Neuhaus, E., Sullivan C.A.W., Bernier, R.A., Bookheimer, S.Y., Dapretto, M., Geschwind, D.H., Jack, A., McPartland, J.C., van Horn, J.D., Pelphrey, K.A., Gupta, A.R., Webb, S.J., & the ACE GENDAAR Network. (2025). A common genetic variant in the Neurexin family member *CNTNAP2* is related to language but not communication skills in youth with Autism Spectrum Disorder. *Autism Research, 18,* 898–908.
13. Minnigulova, A., Dragoy, O., & **Arutiunian, V.** (2025). Atypical segregation of frontoparietal and sensorimotor networks is related to social and executive function impairments in children with ASD. *Brain Imaging and Behavior, 19,* 846–855.
14. Minnigulova, A., Karpychev, V., Davydova, E., Pereverzeva, D., Sorokin, A., Tyushkevich, S., Mamokhina, U., Danilina, K., Dragoy, O., & **Arutiunian, V.** (2025). Altered thalamotemporal structural connectivity is associated with autistic traits in children with ASD. *Behavioural Brain Research, 481,* 115414.
15. Samoylov, I., Bolgina, T., Lonshakov, G., Gomozova, M., **Arutiunian, V.,** & Dragoy, O. (2025). Patterns of cortical activity in a silent single-word reading task depend on word frequency and age-related differences: an MEG study. *Psychophysiology, 62,* e70192.
16. Protopova, M., Bolgina, T., **Arutiunian, V.,** Dragoy, O. (2025). Language Localization from Magnetoencephalography (MEG) Beta-Power Dynamics During Sentence Completion. *European Journal of Neuroscience, 62,* e70282.
17. Samoylov, I., Arcara, G., Buyanova, I., Davydova, E., Pereverzeva, D., Sorokin, A., Tyushkevich, S., Mamokhina, U., Danilina, K., Dragoy, O., & **Arutiunian, V.** (2024). Altered neural synchronization in response to 2 Hz amplitude-modulated tones in the auditory cortex of children with Autism Spectrum Disorder: An MEG study. *International Journal of Psychophysiology, 203,* 112405.
18. **Arutiunian, V.**, Arcara, G., Buyanova, I., Fedorov, M., Davydova, E., Pereverzeva, D., Sorokin, A., Tyushkevich, S., Mamokhina, U., Danilina, K., & Dragoy, O. (2024). Abnormalities in both stimulus-induced and baseline MEG alpha oscillations in the auditory cortex of children with Autism Spectrum Disorder. *Brain Structure and Function, 229,* 1225–1242.
19. **Arutiunian, V.**, Santhosh, M., Neuhaus, E., Borland, H., Tompkins, C., Bernier, R.A., Bookheimer, S.Y., Dapretto, M., Gupta, A.R., Jack, A., Jeste, S., McPartland, J.C., Naples, A., van Horn, J.D., Pelphrey, K.A., & Webb, S.J. (2024). The relationship between gamma-band neural oscillations and language skills in youth with Autism Spectrum Disorder and their first-degree relatives. *Molecular Autism, 15,* 19.
20. **Arutiunian, V.**, Arcara, G., Buyanova, I., Buivolova, O., Davydova, E., Pereverzeva, D., Sorokin, A., Tyushkevich, S., Mamokhina, U., Danilina, K., & Dragoy, O. (2023). Event-Related Desynchronization of MEG Alpha-

Band Oscillations During Simultaneous Presentation of Audio and Visual Stimuli in Children with Autism Spectrum Disorder. *Brain Sciences, 13,* 1313.

21. **Arutiunian, V.**, Davydova, E., Pereverzeva, D., Sorokin, A., Tyushkevich, S., Mamokhina, U., Danilina, K., & Dragoy, O. (2023). Reduced gray matter volume of amygdala and hippocampus is associated with the severity of autistic symptoms and language abilities in school-aged children with Autism Spectrum Disorder: an exploratory study. *Brain Structure and Function, 228,* 1573–1579.
22. Minnigulova, A., Davydova, E., Pereverzeva, D., Sorokin, A., Tyushkevich, S., Mamokhina, U., Danilina, K., Dragoy, O., & **Arutiunian, V.** (2023). Corpus callosum organization and its implication to core and co-occurring symptoms of Autism Spectrum Disorder. *Brain Structure and Function, 228,* 775–785.
23. **Arutiunian, V.**, Arcara, G., Buyanova, I., Davydova, E., Pereverzeva, D., Sorokin, A., Tyushkevich, S., Mamokhina, U., Danilina, K., & Dragoy, O. (2023). Neuromagnetic 40 Hz Auditory Steady-State Response in the left auditory cortex is related to language comprehension in children with Autism Spectrum Disorder. *Progress in Neuropsychopharmacology and Biological Psychiatry, 122,* 110690.
24. **Arutiunian, V.**, Gomozova, M., Minnigulova, A., Davydova, E., Pereverzeva, D., Sorokin, A., Tyushkevich, S., Mamokhina, U., Danilina, K., & Dragoy, O. (2023). Structural brain abnormalities and their association with language impairment in school-aged children with Autism Spectrum Disorder. *Scientific Reports, 13,* 1172.
25. Yurchenko, A., **Arutiunian, V.**, Shitova, N.M., Bergelson, M., & Dragoy, O. (2023). Registered switching involving lexical-semantic processing in Russian: An ERP study. *Journal of Neurolinguistics, 65,* 101111.
26. **Arutiunian, V.**, Arcara, G., Buyanova, I., Gomozova, M., & Dragoy, O. (2022). The age-related changes in 40 Hz Auditory Steady-State Response and sustained Event-Related Fields to the same amplitude-modulated tones in typically developing children: A magnetoencephalography study. *Human Brain Mapping, 43,* 5370–5383.
27. **Arutiunian, V.**, Lopukhina, A., Minnigulova, A., Shlyakhova, A., Davydova, E., Pereverzeva, D., Sorokin, A., Tyushkevich, S., Mamokhina, U., Danilina, K., & Dragoy, O. (2022). Language Abilities of Russian Primary-School-Aged Children with Autism Spectrum Disorder: Evidence from Comprehensive Assessment. *Journal of Autism and Developmental Disorders, 52,* 584–599.
28. **Arutiunian, V.**, Lopukhina, A., Minnigulova, A., Shlyakhova, A., Davydova, E., Pereverzeva, D., Sorokin, A., Tyushkevich, S., Mamokhina, U., Danilina, K., & Dragoy, O. (2021). Expressive and Receptive Language in Russian Primary-School-Aged Children with Autism Spectrum Disorder. *Research in Developmental Disabilities, 117,* 104042.
29. **Arutiunian, V.**, & Lopukhina, A. (2020). The effects of phonological neighborhood density in childhood word production and recognition in Russian are opposite to English. *Journal of Child Language, 47*(6), 1244–1262.

Articles in international peer-reviewed journals (under review)

30. **Arutiunian, V.,** Davoudi, S., Buckser R., Knoth, I.S., Thurman, A.J., Bélanger, A-M., Martin, C-O., Proteau-Lemieux, M., Tassone, F., Santos, E., Ponzini, M., Rubanarayana, K., Pham, T., Whitlock, K., Bennett, J., Kim, K., Biag, H., Hessl, D.R., Schneider, A., Abbeduto, L., Jacquemont, S., Bolduc, F.V., Hagerman, R., & Lippé, S. (under review). Metformin Reshapes Auditory Neural Dynamics Associated with Language Improvement in Fragile X Syndrome.
31. **Arutiunian, V.,** Davoudi, S., Knoth, I.S., Coutu-Nadeau, E., Desjardins, J., O'Reilly, C., Hosein, A., Gomez, G.B., Martin, C-O., Fontaine, V.K., Bélanger, A.M., Hébert-Lalonde, N., Rouleau, G., D'Arc, B.F., Elsabbagh, M., Jacquemont, S., & Lippé, S. (under review). Auditory Sustained Brain Activity Links Genomic Variation to Language in Neurodevelopmental Disorders.
32. **Arutiunian, V.**, Davoudi, S., Knoth, I.S., Bellemare, S., Deguire, F., Lopez-Arango, G., & Lippé, S. (under review). Face processing theta-band oscillations determine opposing trial-to-trial speech processing theta dynamics during a multisensory task in early infancy: A longitudinal study.
33. **Arutiunian, V.**, Santhosh, M., Tompkins, C., Macdonald, K., Corrigan, S., Dommer K., Shic, F., & Webb, S.J. (under review). EEG gamma power discriminates infants with elevated-likelihood risk for developing autism at the first year of age and associated with later adaptive skills.
34. Coutu-Nadeau, E., **Arutiunian, V.,** Martin, C-O., Fontaine, V.K., Bélanger, A.M., Jizi, K., Knoth, I.S., Hosein, A., Davoudi, S., Scorah, J., Ernst, C., Carter, M., Tardif, C.L., Couture, M., Abadie, P., Joober, R., Hébert-

Lalonde, N., Rouleau, G., Elsabbagh, M., D'Arc, B.F., Jacquemont, S., & Lippé, S. (under review). Behavioral Clustering and Resting-State EEG Signatures of Transdiagnostic Neurodevelopmental Profiles.

35. Opdahl, M., **Arutiunian, V.,** Santhosh, M., Corrigan, S., Dommer, K., Macari, S., Chawarska, K., Schic, F., & Webb, S.J. (under review). Distinct developmental trajectories of periodic and aperiodic EEG during live social interaction in infants at typical and atypical elevated likelihood for neurodevelopmental conditions.
36. Buckser R., Davoudi, S., **Arutiunian, V.,** Knoth, I.S., Thurman, A.J., Bélanger, A-M., Martin, C-O., Proteau-Lemieux, M., Santos, E., Ponzini, M., Rubanarayana, K., Pham, T., Whitlock, K., Bennett, J., Kim, K., Biag, H., Hessl, D.R., Schneider, A., Abbeduto, L., Jacquemont, S., Hagerman, R., Bolduc, F.V., & Lippé, S. (under review). Resting state-EEG Reveals Target Engagement and Treatment Response to Metformin in Fragile X Syndrome and Associates with Improvement in Adaptive Behaviors.
37. Coleman, C.R, Nance, M.G., Jacokes, Z., Druzgal, T.J., **Arutiunian, V.,** Kresse, A., Sullivan, C.A.W., Santhosh, M., Neuhaus, E., Borland, H., Bernier, R.A., Bookheimer, S.Y., Dapretto, M., Jack, A., Jeste, S., McPartland, J.C., Naples, A., Geschwind, D., Gupta, A.R., Webb, S.J., Pelphrey, K.A., Van Horn, J.D., Newman, B.T., & Puglia, M.H., on behalf of the ACE GENDAAR Consortium (under review). Structural Determinants of Signal Speed: A Multimodal Investigation of Face Processing in Autism Spectrum Disorder.
38. Kennedy, G., Baran-Goldwax, M., Pomerleau, R., Chabibi, N., Caplette, L., **Arutiunian, V.,** Knoth, I.S., Jodoin, A.M., & Lippé, S. (under review). Menstrual Cycle Timing Predicts Neural Synchrony in the 40Hz Auditory Steady-State Response: Pilot Evidence for Age-Related Variation.

Conferences (selected)

1. **Arutiunian, V.,** Davoudi, S., Knoth, I.S., Coutu-Nadeau, E., Desjardins, J., O'Reilly, C., Hosein, A., Gomez, G.B., Martin, C-O., Fontaine, V.K., Bélanger, A.M., Hébert-Lalonde, N., Rouleau, G., D'Arc, B.F., Elsabbagh, M., Jacquemont, S., & Lippé, S. EEG-based auditory sustained potential links genomic variation to language functioning in individuals with neurodevelopmental disorders. Poster presented at the Annual Meeting of the Society for Psychophysiological Research, Halifax, NS, Canada, 14 – 17 October 2026.
2. **Arutiunian, V.** *Neural and genetic profiles in subgroups of children with autism.* Invited talk given at the 13th International Conference 'Autism. Challenges and Solutions', Moscow, Russia, 12 – 14 September 2025.
3. **Arutiunian, V.** *EEG as a predictor of Neurodevelopmental Outcomes in Infants with Low Birth Weight or Family History of Autism.* Talk given at the International Society for Autism Research (INSAR) Annual Meeting, Seattle, WA, USA, 30 April – 3 May 2025.
4. **Arutiunian, V.**, Santhosh, M., Macdonald, K., Tompkins, C., Corrigan, S., Shic, F., & Webb, S.J. *Neural and communication profiles in infants at elevated risk for developing ASD and DD.* Poster presented at the Annual Meeting of the Flux Society (Society for Developmental Cognitive Neuroscience), Baltimore, MD, USA, 27 – 30 September 2024.
5. **Arutiunian, V.** *Neurophysiological mechanisms of language impairment in children with Autism Spectrum Disorder*. Invited talk given at the 11th International Summer Neurolinguistic School, Moscow, Russia, 3 – 5 July 2024.
6. **Arutiunian, V.** *Gamma-band neural activity and its relation to language skills and core symptoms of Autism Spectrum Disorder*. Invited talk given at the 12th International Conference 'Autism. Challenges and Solutions', Abu Dhabi, United Arab Emirates, 27 – 30 April 2024.
7. **Arutiunian, V.**, Santhosh, M., Corrigan, S., Pelphrey, K., Jeste, S., & Webb, S.J. *Language impairment in children with Autism Spectrum Disorder and their first-degree relatives related to gamma-band neural activity in EEG.* Poster presented at the Annual Meeting of the Society for Psychophysiological Research, New Orleans, LA, USA, 27 September – 1 October 2023.
8. **Arutiunian, V.** *A common genetic variant in the Neurexin family member CNTNAP2 is related to structural language skills in youth with Autism Spectrum Disorder*. Invited talk given at the 11th International Conference 'Autism. Challenges and Solutions', Abu Dhabi, United Arab Emirates, 28 – 30 April 2023.
9. **Arutiunian, V.**, Santhosh, M., Corrigan, S., Pelphrey, K., & Webb, S.J. *Language abilities of unaffected siblings of youth with ASD are lower in comparison to typically developing controls as revealed with CELF*. Poster presented at the Meeting on Language in Autism, Durham, NC, USA, 9 – 11 March 2023.

10. **Arutiunian, V.** *The structural brain abnormalities and their association with language impairment in children with ASD*. Invited talk given at the 10th International Conference 'Autism. Challenges and Solutions', Moscow, Russia, 23 – 25 April 2022.
11. **Arutiunian, V.**, Arcara, G., Gomozova, M., Buyanova, I., Davydova, E., Pereverzeva, D., Sorokin, A., Tyushkevich, S., Mamokhina, U., Danilina, K., & Dragoy, O. *40 Hz ASSR in the left hemisphere is associated with language development in children with Autism Spectrum Disorder: An MEG study*. Poster presented at the MEGUK conference, Cambridge, UK, 6 – 7 September 2021.
12. **Arutiunian, V.**, Lopukhina, A., Minnigulova, A., Shlyakhova, A., Davydova, E., Pereverzeva, D., Sorokin, A., Tyushkevich, S., Mamokhina, U., Danilina, K., & Dragoy, O. *Language Profiles of Russian Primary-School-Aged Children with Autism Spectrum Disorder*. Poster presented at the International Society for Autism Research Annual Meeting (Virtual), Boston, MA, USA, 3 – 7 May 2021.
13. **Arutiunian, V.**, Arcara, G., & Dragoy, O. *The maturation of auditory gamma oscillations reflects cortical excitability: An MEG study*. Poster presented at the Society for Research in Child Development Biennial Meeting (Virtual), Minneapolis, USA, 7–9 April 2021.
14. **Arutiunian, V.**, Arcara, G., Lopukhina, A, & Dragoy, O. *The development of auditory gamma synchrony (40 Hz ASSR) in typically developing children: An MEG study*. Poster presented at the I National Congress on Cognitive Research, Artificial Intelligence, and Neuroinformatics, Moscow, Russia, 10–16 October 2020.
15. **Arutiunian, V.**, Minnigulova, A., Sorokin, A., Davydova, E., Pereverzeva, D., Tyushkevich, S., Mamokhina, U., Danilina, K., & Lopukhina, A. *Expressive and Receptive Vocabulary Impairments in Primary-School-Aged Children with Autism Spectrum Disorder: A Pilot Study in Russian*. Poster presented at the International Society for Autism Research Annual Meeting (Virtual), Seattle, WA, USA, 3 June 2020.
16. **Arutiunian, V.**, Minnigulova, A., & Lopukhina, A. *Nonword repetition is impaired in children with Autism Spectrum Disorder*. Proceedings of the Satellite of AMLaP conference "Typical and Atypical Language Development Symposium", Moscow, Russia, 4 September 2019.
17. **Arutiunian, V.**, & Lopukhina, A. *The influence of phonological neighbourhood density on spoken-word comprehension in Russian children: Evidence from eye-tracking*. Poster presented at the I International conference "ABC: Asia-Pacific BabyLab Constellation", Singapore, 4–5 October 2018.
18. **Arutiunian, V.**, & Lopukhina, A. *Phonological neighbourhood density effect in word production in Russian children: A naming-task study*. Poster presented at the 3rd Summer School "Infant Studies on Language Development in Europe", Potsdam, Germany, 11–15 June 2018.

PROFESSIONAL MEMBERSHIPS

2025 – present	International Society for Autism Research (member)
2025 – present	The Society for Psychophysiological Research (member)
2024 – 2025	Flux: the Society for Developmental Cognitive Neuroscience (member)
2023 – 2024	The Society for Psychophysiological Research (member)

INVITED REVIEWER

Journal of Child Psychology and Psychiatry
Infancy
Molecular Psychiatry
Frontiers in Psychology
Psychiatry and Clinical Neuroscience
Translational Psychiatry
Journal of Neural Engineering
Scientific Reports
Autism Research
Molecular Autism
Journal of Child Language
Brain and Cognition
Human Brain Mapping

TEACHING AND ADVISING

Courses

2025 – present **Neurodiversity** (PSY 3273; BA in Psychology; BA in Cognitive Neuroscience; BA in Psychology – Campus Laval; University of Montreal, QC, Canada). Development of the course for undergraduate programs in psychology and cognitive neuroscience.

2025 – present **EEG data preprocessing and analysis.** I organize workshops and courses at the lab as well as support undergraduate and graduate students on the EEG preprocessing and analysis using standardized pipelines.

2023 – 2025 **Summer Clinical Scholars Program.** I assisted in teaching courses on clinical assessment with the Vineland Adaptive Behavioral Scale.

2023 – 2024 **Seattle Children's Autism Journal Club.** I co-directed a bi-weekly academic meeting for the staff members and interested research community to discuss scientific topics related to language impairment and processing in autism.

2019 – 2022 **MEG / EEG data analysis.** I regularly organized the short courses at the Center for Language and Brain on EEG and MEG data analysis in Brainstorm (Matlab) (basic signal pre-processing, basic event-related potentials / fields analysis, time-frequency analysis, source estimation).

2021 – 2022 **Research Methods in Neuroscience** (BA program; School of Linguistics, Faculty of Humanities; HSE University, Moscow, Russia).

2021 – 2022 **Psycho- and Neurolinguistics;** module *Structural and functional neuroanatomy; neuroimaging methods* (BA program; School of Linguistics, Faculty of Humanities; HSE University, Moscow, Russia).

2021 **EEG methods in neurolinguistics** (summer school for graduate students; Sirius University, Sochi, Russia).

2020 **Practicum in Psycholinguistics** (MSc program; School of Linguistics, Faculty of Humanities; HSE University, Moscow, Russia).

Supervisor of students' theses and projects

2025 – present Emmanuelle Coutu-Nadeau (MSc in Psychology, University of Montreal, Montreal QC, Canada; *Supervisor*):
EEG-based Neurodevelopmental Analysis in ASD/ADHD youth via Learning Algorithms.

2025 – present Gabriela Kennedy (MSc in Psychology, University of Montreal, Montreal QC, Canada; *Supervisor*):
Exploring Sex Differences in EEG-Cortisol Associations During Adolescence and Young Adulthood.

2023 – present Ilya Samoylov (PhD in Cognitive Neuroscience, HSE University, Moscow, Russia; *Academic Supervisor*):
Neural Mechanisms of Language Processing in Children with Neurodevelopmental Disorders.

2022 – 2026 Alina Minnigulova (PhD in Cognitive Neuroscience, HSE University, Moscow, Russia; *Academic Supervisor*):
Neural signatures of core and co-occurring symptoms in Autism Spectrum Disorder: an MRI-based connectivity study.

2024 – 2025 Morgan Opdahl (SURFiN fellow/intern, Seattle Children's & University of Washington, Seattle, USA): *Excitation/inhibition imbalance, its neural correlates and relation to clinical phenotypes in youth with Autism Spectrum Disorder.*

2023 – 2024 Xinyue Yu (SURFiN fellow/intern, Seattle Children's & University of Washington, Seattle, USA):
Early predictors of language impairment in infants at risk for developing autism.

2023 – 2024 Aya Sahib (SURFiN fellow/intern, Seattle Children's & University of Washington, Seattle, USA):
Early predictors of language impairment in infants at risk for developing autism.

2020 – 2022 Alina Minnigulova (2022, MSc in Cognitive Neuroscience, HSE University, Moscow, Russia):
Auditory processing during visual attention in typically developing children and children with Autism Spectrum Disorder: An ERP study.

2018 – 2022 Semen Kudriavtsev (2022, BA in Linguistics, HSE University, Moscow, Russia): *Neural habituation as an adaptation mechanism to repetitive sounds in typically developing children and children with Autism Spectrum Disorder: An ERP study.*

2020 Alina Minnigulova (2020, BA in Linguistics, HSE University, Nizhniy Novgorod, Russia): *Phonological processing deficit in children with Autism Spectrum Disorder.*

SKILLS

Conducting experiments using behavioral (language assessment tests, eye-tracking), neuroimaging (magnetoencephalography, MEG; electroencephalography, EEG; voxel-based and surface-based morphometry, MRI), and genetic methods.

- **MEG:** *Data collection:* 306-channel Vectorview, Electa Neuromag system; *Basic pre-processing:* MaxFilter software; *Neuronal data analysis:* Brainstorm toolbox (Matlab) and MNE Python toolbox (Python) – event-related fields (ERF) analysis, time-frequency analysis, source estimation.
- **EEG:** *Data collection:* 64/128-channel ANT system, EEGO software; 128-channel Net Amps 300 system with HydroCel nets, Net Station software; *Neuronal data analysis:* Brainstorm toolbox (Matlab) and MNE Python toolbox (Python) – event-related potentials (ERP) analysis, time-frequency analysis, source estimation; BEAPP-HAPPEE toolbox (Matlab) – frequency analysis.
- **MRI:** *Segmentation:* FreeSurfer software, SPM / CAT12 toolboxes (Matlab); *Voxel-based and Surface-based morphometry, ROI analysis:* SPM / CAT12 toolboxes (Matlab); *Statistical design and analysis:* SPM / CAT12 toolboxes (Matlab).
- *Eye-tracking data collection:* SMI RED-m, SMI Experiment Center.
- Statistics: R (frequentist statistics, mixed effects models, mediation analysis, cluster-based analysis).
- **Genetics:** gene co-expression analysis, molecular pathway and ontology analysis, transcriptome analysis.